



Understanding the Influence of Convective Weather on Ozone Pollution in Oklahoma City, OK

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i) Background

- Ozone (O₃) is a result of the photolysis of oxygen molecules
- NO_x and VOCs contribute to O₃ production and are emitted from several anthropogenic and biogenic sources
- O₃ production is accelerated on summer days with high temperatures
- O₃ poses several negative health impacts, particularly to those with pre-existing conditions
- O₃ production on severe convective weather days in Oklahoma City, Oklahoma remains unclear**

ii) Goals

- Identify whether O₃ concentrations are statistically different on convective severe weather days compared to non-convective severe weather days
- Identify methods to quantify such significance
- Further develop my real-world, applied research problem-solving skills and critical thinking
- Develop the confidence to engage in the research process from proposal to presentation

iii) Identifying Storm Days

Local Significant Weather Events

[Weather.gov](#) > [Norman, OK](#) > Local Significant Weather Events

[Current Hazards](#) [Current Conditions](#) [Radar](#) [Forecasts](#) [Rivers and Lakes](#) [Climate and](#)

Storm Data

Tornado Data

Climate Data

2020s

2010-2019

2000-2009

1990-1999

1980-1989

Significant Weather Events That Occurred from 2020-2029

Fig. 2: Local Significant Weather Events Database from National Weather Service Norman

Case Study Dates Identified:

2024 May 1 to May 5
2024 April 22 to May 3
2023 May 6 to May 16
2023 April 14 to April 24
2022 April 18 to April 28
2021 October 8 to October 18

iv) Ozone Time Series

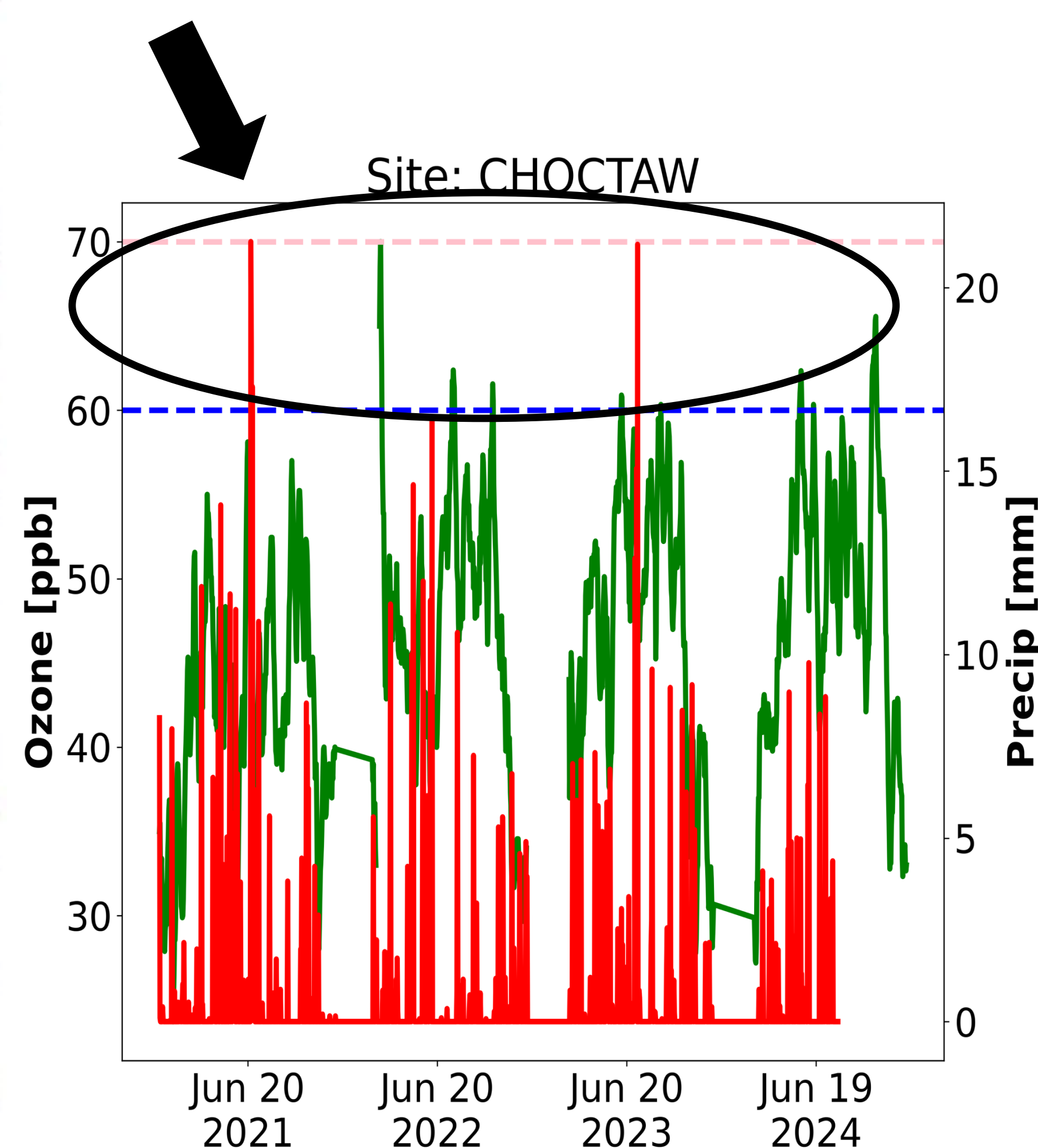


Fig. 3: Two distinct ozone pollution events in 2021 and 2023
Red: Ozone. Green: Precipitation

iv) Ozone Time Series cont.

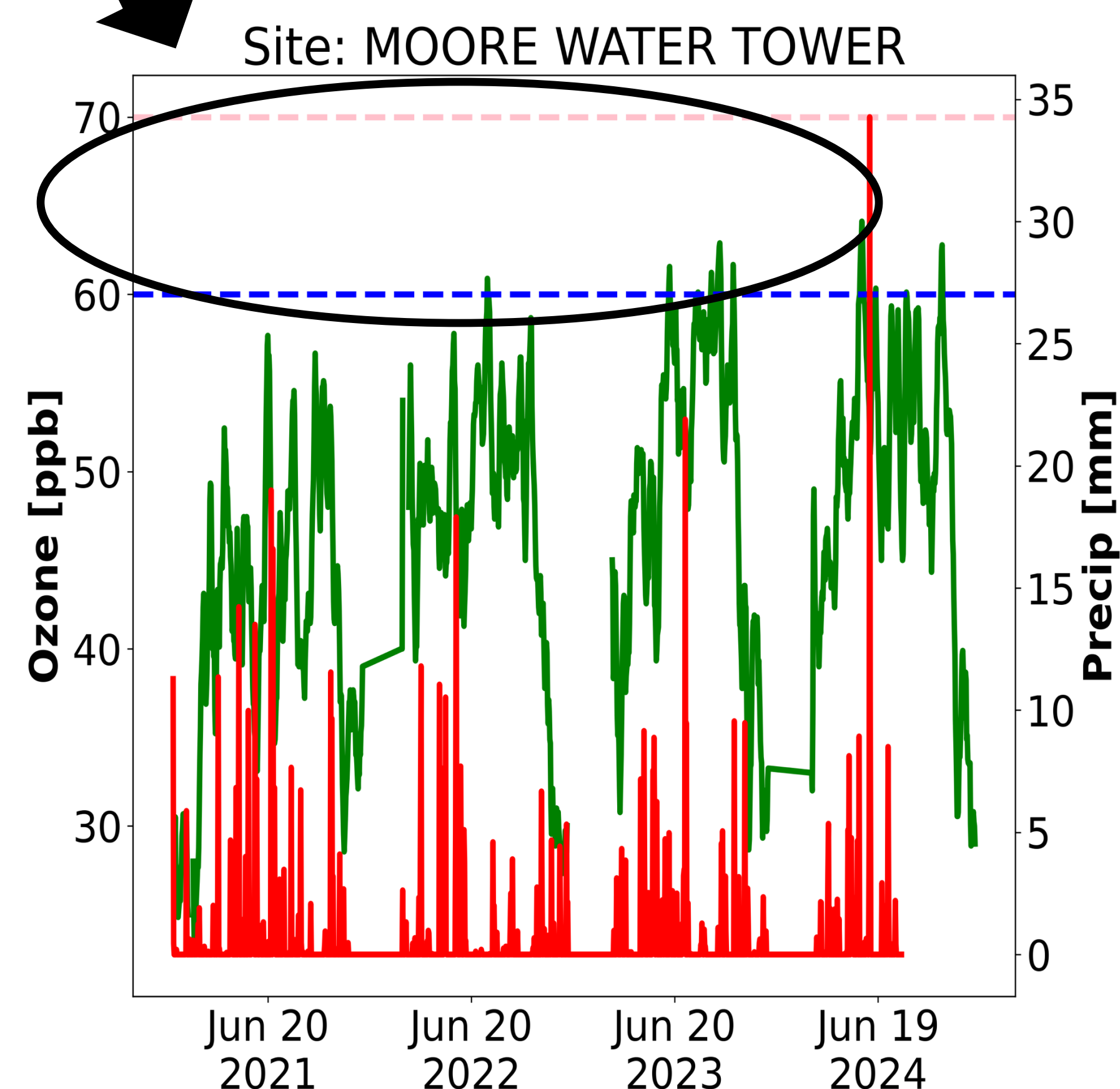


Fig. 4: 2021 and 2023 ozone pollution events are **NOT** seen in Moore

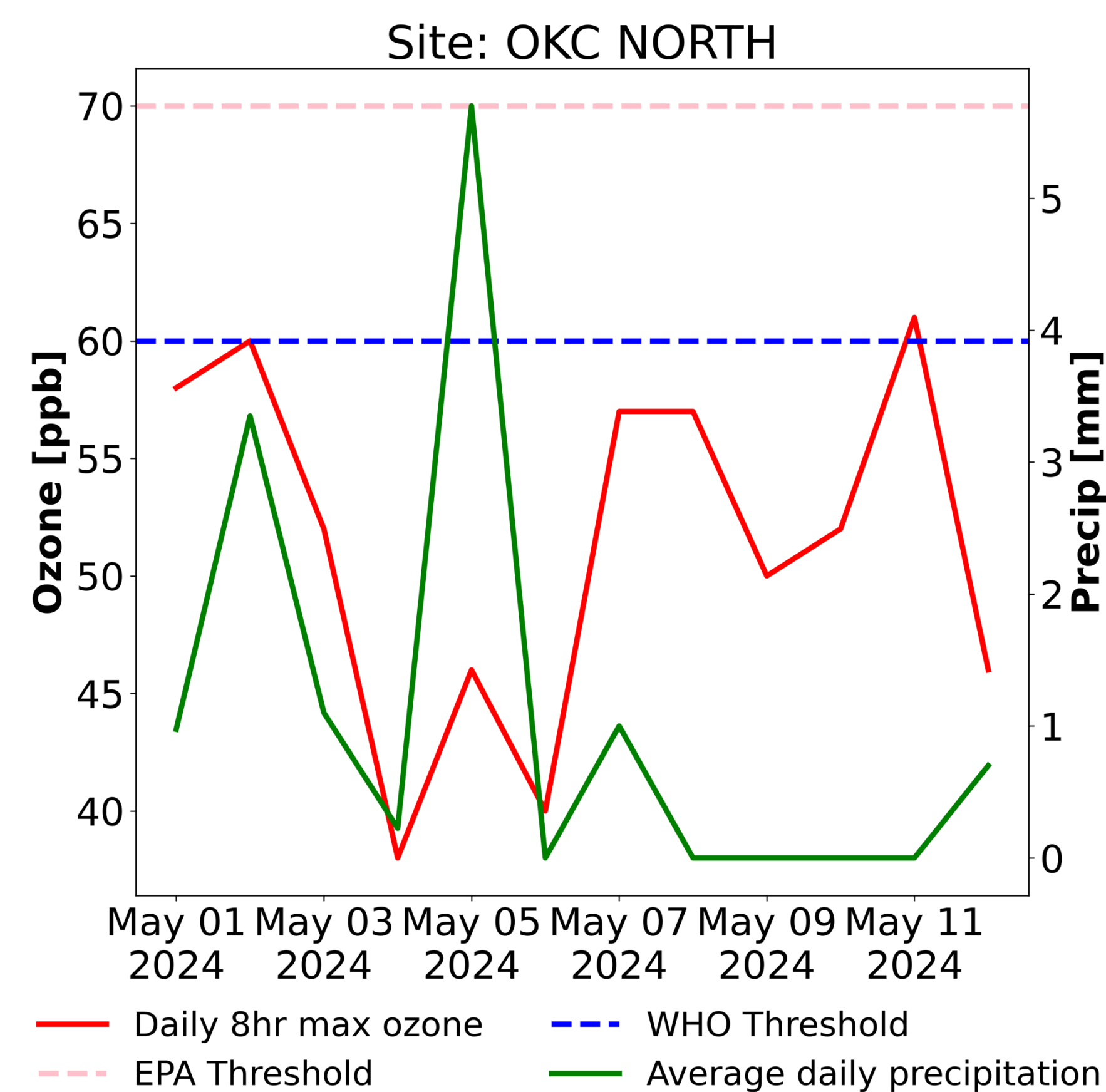


Fig. 6: Similarly timed ozone pollution events (e.g. fig. 5). Ozone level was lower following larger precipitation amounts

Meteorological variables have a crucial role in both the formation AND MOVEMENT OF OZONE

v) Key Points

- Meteorological factors play a crucial role in the advection and formation of ozone
- Northern sites in OKC exhibited **higher ozone pollution concentrations**
- Rainfall markedly decreased ozone concentrations
- Ozone data availability particularly **limits** our analysis
- Hourly ozone measurements would provide crucial insights that would enable us to more directly answer our original research question

viii) References

- Hu, X.-M., Xue, M., Kong, F., & Zhang, H. (2019). Meteorological Conditions During an Ozone Episode in Dallas-Fort Worth, Texas, and Impact of Their Modeling Uncertainties on Air Quality Prediction. *Journal of Geophysical Research: Atmospheres*, 124(4), 1941–1961. <https://doi.org/10.1029/2018JD029791>
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ix) Acknowledgements

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Fig. 1: Ozone monitoring sites across Oklahoma City